



Pressure dependent kinetic analysis of pathways to naphthalene from cyclopentadienyl recombination



Alan E. Long^a, Shamel S. Merchant^b, Aäron G. Vandeputte^c, Hans-Heinrich Carstensen^d, Alexander J. Vervust^d, Guy B. Marin^d, Kevin M. Van Geem^d, William H. Green^{a,*}

^a Department of Chemical Engineering, Massachusetts Institute of Technology, 77 Massachusetts Ave., Cambridge, MA 02139, USA

^b ExxonMobil Research and Engineering, 1545 Route US 22 East, Annandale, N.J. 08801, USA

^c SABIC, Geleen Technology Center, Urmonderbaan 22, 6167 RD Geleen, The Netherlands

^d Laboratory for Chemical Technology, Ghent University, Technologiepark 914, B-9052 Gent, Belgium

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ABSTRACT

Cyclopentadiene (CPD) and cyclopentadienyl radical (CPDyl) reactions are known to provide fast routes to naphthalene and other polycyclic aromatic hydrocarbon (PAH) precursors in many systems. In this work, we combine literature quantum chemical pathways for the CPDyl + CPDyl recombination reaction and provide pressure dependent rate coefficient calculations and analysis. We find that the simplified 1-step global reaction leading to naphthalene and two H atoms used in many kinetic models is not an adequate description of this chemistry at conditions of relevance to pyrolysis and steam cracking. The C₁₀H₁₀ species is observed to live long enough to undergo H abstraction reactions to enter the C₁₀H₉ potential energy surface (PES). Rate coefficient expressions as functions of T and P are reported in CHEMKIN format for future use in kinetic modeling.

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1. Introduction

Polycyclic aromatic hydrocarbons (PAHs) are a class of molecules comprised of hundreds of chemicals produced from various anthropogenic sources, such as the incomplete combustion of fuels in boilers and heating devices, oil refining processes, and the combustion of transportation fuels (in particular diesel). Many PAHs are known to be carcinogenic or mutagenic as well as important precursors to soot [1–5]. Increasing energy demands and more stringent emission regulations have spurred research into various new technologies that reduce PAH emissions [6–8]. Despite the attention that these PAHs have attracted, many questions related to their formation remain not fully understood, i.e. what are the key reactions responsible for the formation of PAHs and what are the main PAH precursors? Researchers have developed extensive kinetic mechanisms describing low and high temperature fuel oxidation, but few exist that are able to predict the PAH growth and subsequent soot formation [9–11]. Some key efforts toward the latter have come from Frenklach, Appel et al., and Richter and Howard [12–15]. However, these studies focused primarily on the chemistry at high temperature flame conditions, at which the HACA mechanisms first proposed by Frenklach in 1991 as well var-

ious similar pathways summarized in the review article by Richter and Howard [12,14] dominate. The radicals involved in these reactions are highly reactive or energetic and thus not present in sufficient concentrations at lower temperatures. Given their increased stability, resonantly stabilized radicals such as allyl, propargyl and cyclopentadienyl (CPDyl) have been suggested as possible precursors of aromatic products at low or moderate temperatures [16–19]. However, most of these reactions lead to mono-aromatic hydrocarbons. Several experimental pyrolysis studies at lower temperatures detected CPD as an important product and Melton et al. showed that PAH formation was most sensitive to the CPD concentration [20–23]. Since CPD is easily converted to CPDyl this suggests that CPDyl radicals play a crucial role in PAH formation. In this work, the focus is on the development of a detailed kinetic network for cyclopentadienyl recombination that is relevant at pyrolytic and low-temperature combustion conditions to describe the formation of naphthalene, the simplest PAH and an important PAH precursor.

As briefly indicated above, a number of researchers have explored the thermal decomposition of CPD experimentally. In an attempt to obtain bond dissociation energies, the decomposition of CPD under pyrolytic conditions was studied by Szwarc in 1950 [24]. This led to a complicated decomposition spectrum, containing H₂, CH₄, C₂ hydrocarbons and other species hinting at the complexity of the CPD reaction network. In 1972 Spielman and Cramers

* Corresponding author.

E-mail address: whgreen@mit.edu (W.H. Green).

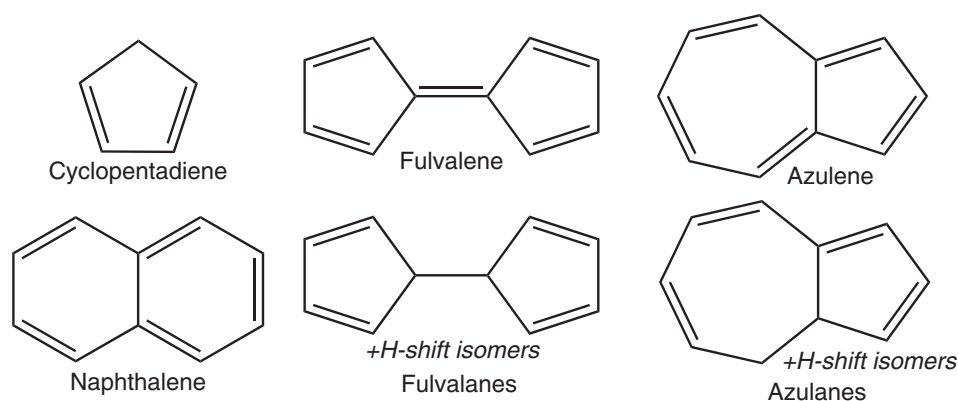


Fig. 1. Primary structures of this study. An ending of -yl added to a species indicates a radical of that species formed by hydrogen loss.

first proposed a potential role of CPD in the formation of the initial aromatics, observing products such as styrene, indene, and naphthalene under pyrolytic conditions [25]. This was then validated by studies of the pyrolysis and hydrogenolysis of phenol [26–28]. Burcat et al. studied the high-temperature decomposition reactions of CPD in a shock-tube in 1996, and modeled it with a simple model of 36 reactions. Roy et al. also studied CPD reactions using shock tube experiments and reported rate coefficients for the C–H bond scission and the reaction between CPD and a hydrogen atom [29,30]. While the Spielman study and others showed qualitatively that naphthalene appears to be produced from CPD, in 2003, Murakami et al. first experimentally derived a rate expression for a net reaction from CPDyl to naphthalene, assuming CPD initially forms its relatively stable radical [31]. More recently, experimental efforts to understand CPD pyrolysis chemistry have shifted toward flow reactor studies [32,33]. Djokic et al. and Kim et al. identified and quantified numerous pyrolytic products from CPD up to anthracene, phenanthrene and fluorene supporting the important role of CPD and CPDyl in PAH formation. The crucial role of CPDyl radicals in PAH formation is additionally confirmed by pyrolysis studies of anisole (which forms CPDyl by methyl loss and carbon monoxide ejection) and dimethylfuran [20,34,35]. More recently, in 2015, Knyazev and Popov experimentally investigated the total self-reaction rate of CPDyl, finding it to show significantly faster kinetics when compared with other similar radical self-reactions. Naphthalene and Azulene were detected as final products with the former being the major product [36].

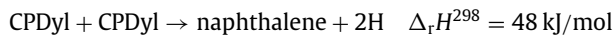
The first quantum chemical exploration of the potential energy surface (PES) for CPDyl recombination was reported by Melius et al. in 1996. This study used the BAC-MP4 and MP2 levels of theory to characterize pathways in which two CPDyl radicals recombine, subsequently lose an H atom to form a fulvalanyl radical (Fig. 1), which then isomerizes and ejects an additional H atom to yield naphthalene. The importance of hydrogen atom mobility for PAH formation was emphasized by this work as was the role of resonantly stabilized intermediates [37]. A 2006 paper by Wang et al. extended the surface using the B3LYP level to incorporate pathways to benzene and indene as well as naphthalene. However, that work focused on initial reactions between CPD and CPDyl rather than two CPDyl radicals. It was observed that C–C β scission routes tended to be favored at high temperatures [38]. A 2007 paper by Kislov and Mebel used the RCCSD(T)/6-311G(d,p) level to investigate the $C_{10}H_9$ potential energy surface. Routes to naphthalene, azulene, and fulvalene were compared. Assuming unimolecular reactions in the high pressure limit, fulvalene was computed to be the dominant product above 1500 K, while naphthalene was the primary species at lower temperatures, and azulene was always a minor product. Kislov and Mebel extended their work in 2008

to incorporate the relevant portion of the $C_{10}H_{11}$ surface consisting principally of routes to indene calculated at the G3 level [39]. Subsequent work by Kislov and Mebel elucidated the poor likelihood of reaction pathways that would produce molecular hydrogen along the way to naphthalene from two CPDyl radicals as had been proposed as a global step for the pathway and incorporated into some kinetic mechanisms [40,41]. A 2012 paper by Cavallotti et al. revisited routes from the reaction of CPD with CPDyl using the B3LYP level for most calculations and ROCBS-QB3 for important flux determining steps along the pathways analyzed. Additional routes were located for the formation of indene, benzene, vinylfulvene, and phenyl butadiene [42]. In 2013 Cavallotti et al. analyzed the $C_{10}H_{10}$ surface at the CBS-QB3 level, focusing on the formation of the fulvalanyl and azulanyl radicals. Routes to the azulanyl radical were reported to dominate up to 1450 K [43]. This result in turn added relevance of the spiran and methyl walk pathways from azulene to naphthalene shown by Alder et al. in 2003 [44]. Some additional discussion of many of these quantum chemical studies can be found in a recent paper by Mebel et al. [45].

Rate coefficients are needed to quantitatively model PAH formation kinetics. A number of the previously noted studies presented high pressure limit rate coefficients and global step rate constants. Wang et al. reported only high-pressure limit Arrhenius expressions for the primary reaction pathway branching reactions they studied [38]. The 2007 and 2008 Kislov and Mebel papers reported rate coefficients and equilibrium constants for all important reactions they located. These were reported at various relevant temperatures in table format [39,46]. Global overall Arrhenius expressions were reported by Cavallotti et al. in 2012 and 2013 for the formation of the terminal species of interest in those studies [42,43]. However, none of these studies incorporated pressure dependence in their reported rate expressions. While the C_{10} species of interest are relatively large molecules, they are not large enough to be sure their unimolecular reactions are at the high pressure limit at pyrolysis and combustion conditions. This follows from a 2003 paper by Wong et al. which developed convenient criteria for determining the relevance of pressure dependent kinetics for a system [47]. Following this criteria, a recent model of propene pyrolysis includes a set of pressure dependent rate expressions for the progression of CPDyl to naphthalene first described by Melius et al. Rather than calculating the individual rates for the network directly, they are assigned based on analogy to similar reaction whose rate constants are readily available [48]. A brief comparison with this estimated network is presented in the supporting information of this work.

The objective of the current study is two-fold. First it aims to combine the most promising pathways previously studied and use this information to calculate a set of pressure-dependent rate expressions that will be easily used in future kinetic mechanisms.

Specifically, generation of rate expressions in this format is important for database development and improved accuracy of automatic mechanism generating software such as RMG [49]. Secondly, the importance of the direct channel



will be explored. Note that as written, the recombination of CPDyl radicals generates naphthalene in a single step, preventing any bimolecular chemistry of intermediates in this obviously complex reaction sequence. Also, it increases reactivity by converting two low-reactivity radicals into more reactive H atoms.

2. Methodology

2.1. Quantum mechanical calculation of PES

Even though all PES used in this study have already been investigated previously by other groups, the electronic structure calculations are repeated here to obtain a consistent data set for the kinetic analysis. Thermodynamic properties of all species including transition states are calculated using quantum mechanical computations at the CBS-QB3 level as implemented in Gaussian 03 and 09 [50–53]. Bond additivity and spin orbit corrections are included [54,55]. Most vibrations are approximated as harmonic oscillators with frequencies computed at the B3LYP/6-311G(2d,d,p) level, but torsional vibrations of key components are treated as 1D hindered rotors [56]. The calculated CBS-QB3 enthalpy of formation of CPDyl at 298 K is 259 kJ/mol in good agreement with the experimental value [29]. The optimized geometries for all species studied here are readily available in previous PES exploratory studies as referenced.

2.2. Rate coefficient calculations

For pressure dependent calculations, potential energy surfaces for $\text{C}_{10}\text{H}_{10}$ and C_{10}H_9 are modeled using a one-dimensional master equation, accounting for rotational degrees of freedom. Microcanonical rate coefficients are computed using the classical RRKM theory and include Eckart tunneling contributions [57–60]. Densities of states are calculated via inverse Laplace transform of the partition function using the method of steepest descents [61–63]. Phenomenological rate coefficients (which give the dynamics of the total isomer populations $x_i(t)$) are computed from the conventional master equation model (which give the detailed dynamics of the isomer population distributions $p_i(E_n, t)$) using the modified strong collision (MSC) approximation [64,65]. Parameters for the MSC calculation are obtained using the exponential down model $\langle \Delta E_{\text{down}} \rangle$ for the average energy transferred in a collision, which is subsequently converted to the required ΔE_{all} parameter [64]. The average downward energy transfer per collision is calculated according to the following temperature dependent formulation:

$$\langle \Delta E_{\text{down}} \rangle = \langle \Delta E_{\text{down}} \rangle_{300} (T/300 \text{ K})^n \text{ cm}^{-1}$$

with $\langle \Delta E_{\text{down}} \rangle_{300} = 295 \text{ cm}^{-1}$, and $n = 0.7$ for N_2 . These values are adopted based on azulene collisional energy transfer parameters [66]. The collision frequency is computed by assuming a Lennard–Jones potential between the bath gas and the species of interest. Lennard–Jones parameters are approximated by first estimating the critical temperature and pressure using a group additivity method devised by Joback, and then using the equations for a Lennard–Jones gas implemented by Harper et al. in RMG [67–69]. All master equation calculations are performed using the open source CANTHERM software package [70]. Additionally, barrierless hydrogen loss reactions are assumed fast and temperature independent with high pressure limit rate coefficients of

$1 \times 10^{14} \frac{\text{cm}^3}{\text{mol s}}$ in the recombination direction for the purposes of this study. This estimate is in agreement with the 2005 Harding et al. work computing the rates of a variety of hydrogen atom–hydrocarbon radical recombination rates. These rates were shown to vary less than one order of magnitude from the value assumed here. [71]. For the CPDyl + CPDyl association rate, the high pressure limit rate expression from Knyazev and Popov is used. This value was determined experimentally for a range of 300–600 K [36]. Sensitivities for these assumed rates are included in the supporting information.

3. Results and discussion

3.1. $\text{C}_{10}\text{H}_{10}$ PES to fulvalanyl and azulanyl radicals

A schematic of the overall $\text{C}_{10}\text{H}_{10}$ surface examined in this study is shown in Fig. 2. More detailed enthalpy diagrams will be shown in later figures which give the individual structures, but divide the PES into two parts due to the large number of species. When entering the isomerization network through the combination of two CPDyl radicals, the initial adduct has a number of options to further react. First, it can immediately stabilize via collisions and energy transfer to surrounding molecules, thermalizing it to one of the fulvalane species. Based on the energies and barriers, this is expected to be the dominant channel at sufficiently high pressure. Secondly, if the combination of CPDyl radicals produces an adduct with sufficient excess energy, it may be able to rearrange and pass over the high barrier separating the fulvalanes and azulanes before being collision stabilized. Entering the PES at even higher energy levels allows for skipping over the wells of the $\text{C}_{10}\text{H}_{10}$ PES entirely, emitting a hydrogen radical and proceeding immediately to the C_{10}H_9 PES related to fulvalanyl or azulanyl. While a larger degree of atomic rearrangement is necessary to proceed directly to azulanyl, those species are of lower energy than the fulvalanyl isomers. It may thus be possible at certain conditions to generate preferentially azulanyl radicals. Once a fulvalanyl or azulanyl radical has been formed, the C_{10}H_9 species can isomerize or eliminate an H atom to form naphthalene, fulvalene, or azulene, Fig. 7.

3.1.1. Fulvalanyl radical formation

The CPDyl radical recombination reaction and subsequent $\text{C}_{10}\text{H}_{10}$ surface was first examined by the 1996 work by Melius et al., but that work considered only three of the six potential $\text{C}_{10}\text{H}_{10}$ H-shift isomers and neglected one of the three fulvalanyl radical isomers [37]. The complete set of isomers was later shown by the 2013 Cavallotti et al. work [43]. The results of our study show that all wells (isomers) are important and isomerization pathways must be included in the kinetic analysis since missing low-barrier sigmatropic hydrogen shift reactions and isomers will under-predict the total yield of $\text{C}_{10}\text{H}_{10}$ isomers formed in the recombination reaction. A reduced build-up of $\text{C}_{10}\text{H}_{10}$ isomers will lead to an underestimation of naphthalene and other species produced through H loss pathways from these isomers.

The fulvalane/fulvalanyl portion of the network is shown in Fig. 3. The six fulvalane isomers (**T1a** to **T1f**) can easily interconvert due to the relatively low barriers for the sigmatropic H shifts in these components. The CBS-QB3 method predicts barriers that are in the range of 90–120 kJ/mol. This agrees well with previous computational results [37,43]. The isomers with co-planar conjugated rings (**T1c**, **T1d** and **T1e**) are significantly more stable than other isomers. The lowest barrier hydrogen emission occurs between **T1b** and **N1b** at 270 kJ/mol. In general, the C–H bond dissociation energies (BDE) of **T1a** to **T1f** range between 270 and 330 kJ/mol, much smaller than C–H bond energies in alkanes which are around

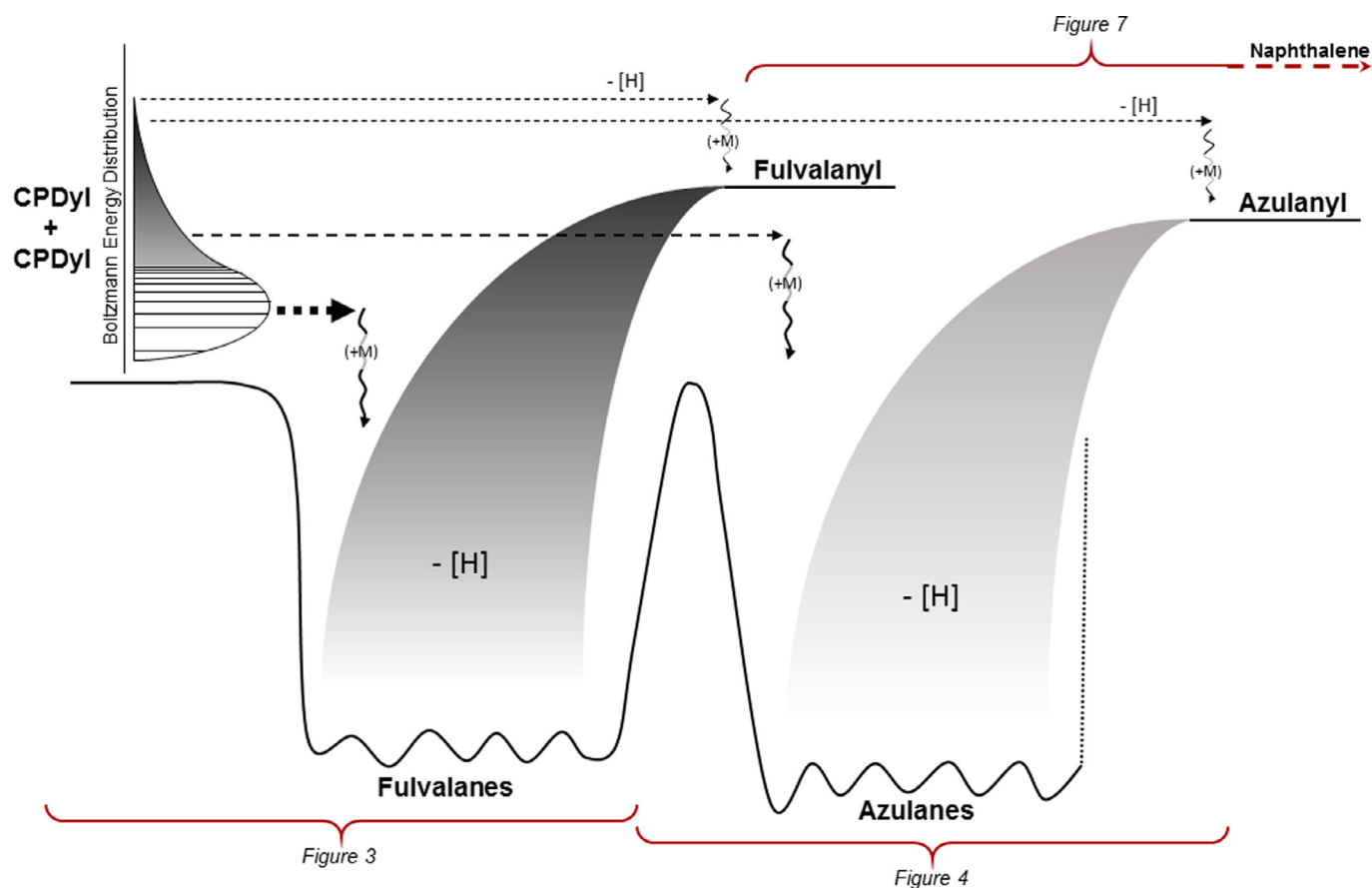


Fig. 2. $C_{10}H_{10}$ surface schematic. The azulanes fall slightly lower in energy than the fulvalanes with an isomerization barrier separating the two groups of roughly equal height to the entrance channel. Due to the number of species, the surface has been broken into several figures as indicated for display purposes, but note that the calculations are performed on the surface as a whole.

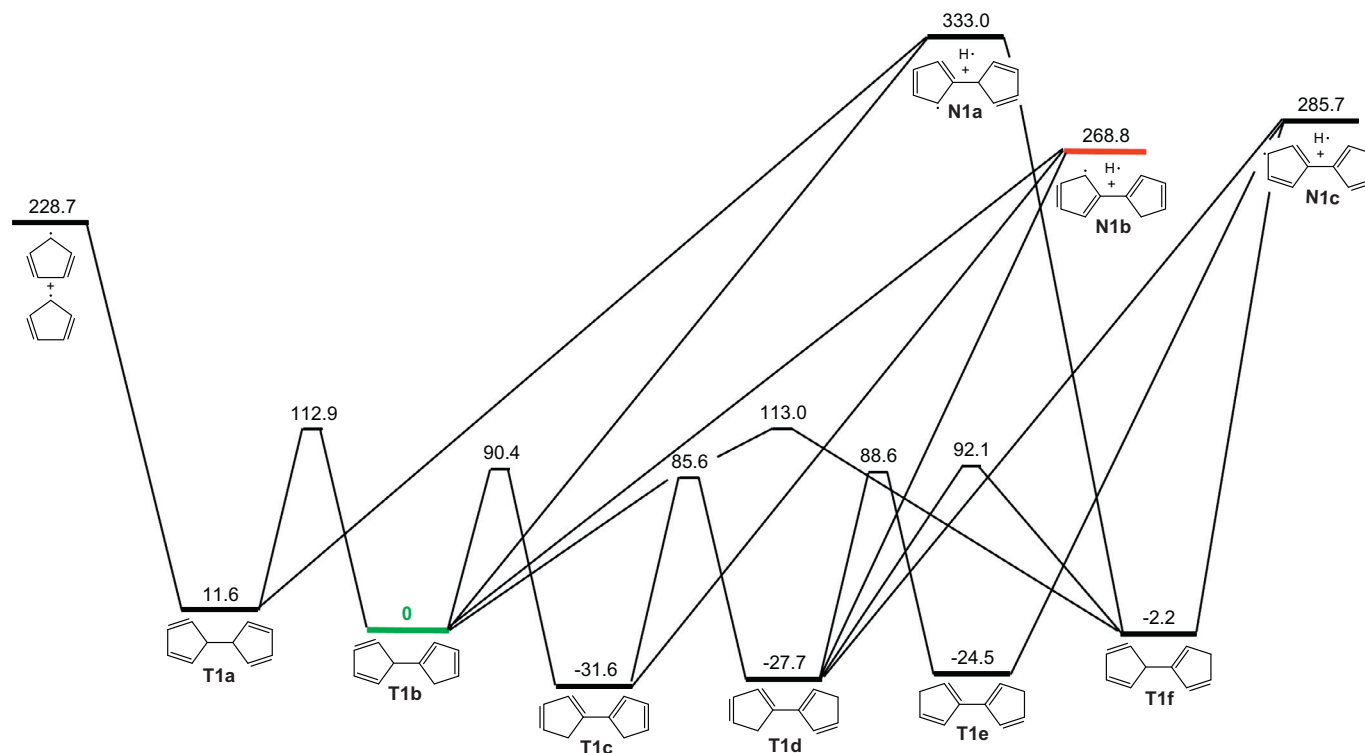


Fig. 3. CBS-QB3 enthalpy diagram $H(0\text{ K})$ (kJ/mol) for the $C_{10}H_{10}$ surface initiated by the recombination of two CPDyl radicals. All enthalpies are relative to the adduct T1b which connects to the other portion of the surface. This network was first reported by Melius et al. and in its complete form by Cavallotti et al. [37,43].

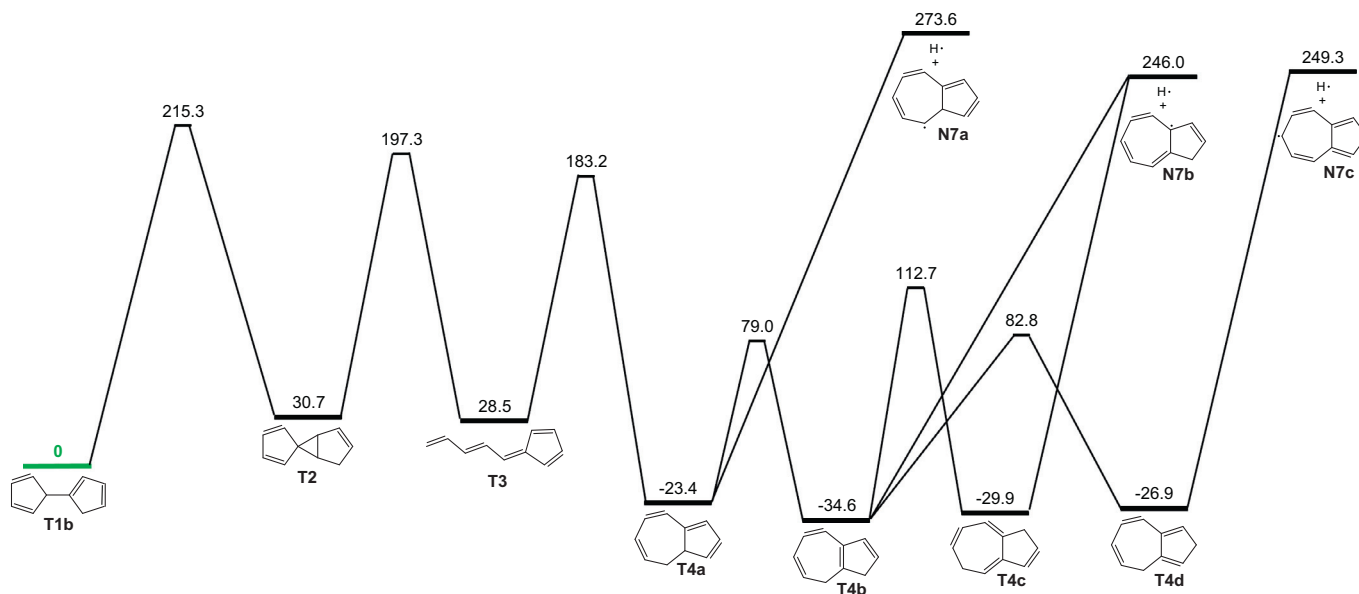


Fig. 4. CBS-QB3 enthalpy diagram $H(0\text{ K})$ (kJ/mol) on the $\text{C}_{10}\text{H}_{10}$ surface leading to the formation of Azulanyl radicals. All enthalpies are relative to component T1b which represents the dominant entrance channel. Pathway first shown by Cavallotti et al. [43]. Additional symmetry allowed azulane species were included in this analysis but are not shown in this figure for simplicity reasons as they are only accessible via a 170 kJ/mol barrier from T4a.

400 kJ/mol and still lower than the allylic C–H bond energy in propene (about 370 kJ/mol) [72].

3.1.2. Azulanyl radical formation

The pathway used for this study (Fig. 4) was first discussed by Cavallotti et al. as being important at lower temperatures [43]. This is primarily due to the increased stability of the azulane species (T4a to T4d). Possible reaction pathways to azulanyl radical were also explored by Kislov and Mebel, though their work examined channels on the C_{10}H_9 surface [39,43].

The portion of the surface relevant for azulanyl radical formation is shown in Fig. 4 and begins with species T1b from Fig. 3. Similar to the formation of fulvalanyl the path is relatively simple, passing through a tricyclic species (T2) followed by a ring opening (T3) and closing to form the azulane species. The azulane species are interconnected by various hydrogen shift reactions. The H-shifts barriers for the azulane species range from around 100–150 kJ/mol indicating a greater resistance to interconversion than is observed for the fulvalanyl species. The lowest energy route to the C_{10}H_9 surface requires an increase in energy of 270 kJ/mol from a fulvalane isomer, about 5 kJ/mol less than any path from an azulane isomer. Though the separating barrier between the fulvalanes and azulanes is relatively high, it is still about 15 kJ/mol lower than the CPDyl + CPDyl entrance channel. Therefore, isomerization to the more stable azulane isomers is energetically feasible, suggesting that rather high azulane concentrations can build up at lower temperatures and thus subsequent azulane pathways need to be included. Three other symmetry allowed azulane species are included in the analysis and calculations for the $\text{C}_{10}\text{H}_{10}$ surface but are omitted from Fig. 4 since they are only accessible by traversing a 170 kJ/mol barrier from T4a.

Some authors have treated this surface as if the chemically activated routes dominate, going even so far as to use a single rate expression to describe the direct conversion from two CPDyl radicals directly to naphthalene and two hydrogen atoms [31]. Here evidence against the validity of this assumption is presented through calculation of the pressure dependent rate constants for this network. The resulting rate coefficients for the various reaction pathways of the recombination of two CPDyl radicals at 1100 K can be found in Fig. 5a as functions of pressure. H atom forming chan-

nels are only of minor importance (<1%) at all pressures investigated. Instead, collision stabilization leads to the formation of various fulvalane isomers. At pressures above atmospheric conditions ($P > 1$ bar), the recombination reaction of the two CPDyl radicals results primarily in the formation of T1a and well-skipping reactions are suppressed. Around 1 bar and below, well-skipping reactions become important and T1c becomes the dominant product. Therefore, at typical conditions the intermediate formed after recombination of two CPDyl radicals only slowly stabilizes, allowing the hydrogen atoms to freely move around in the molecule.

The direct and well-skipping rate coefficients, $k_{A \rightarrow B}(T, P)$, are displayed in matrix form in Fig. 5b. The bottleneck is the conversion of the fulvalanes into species T2, which occurs on a millisecond timescale at 1100 K. Equilibration between the fulvalane species prior to the bottle neck occurs on a much faster microsecond time scale. While slower than the fulvalanes, the azulane species also interconvert rapidly.

With pressure dependent rate analysis confirming that species are stabilized on the $\text{C}_{10}\text{H}_{10}$ surface and do not predominantly proceed directly to C_{10}H_9 , it is thus possible that C_{10}H_9 formation occurs through three different paths: chemically activated hydrogen loss from the CPDyl reaction ($2\text{CPDyl} \rightarrow \text{C}_{10}\text{H}_9 + \text{H}$), hydrogen emission from the $\text{C}_{10}\text{H}_{10}$ species ($\text{C}_{10}\text{H}_{10} \rightarrow \text{C}_{10}\text{H}_9 + \text{H}$), and radical hydrogen abstraction from the $\text{C}_{10}\text{H}_{10}$ species ($\text{C}_{10}\text{H}_{10} + \text{R} \rightarrow \text{C}_{10}\text{H}_9 + \text{RH}$). To see how the three pathways relate to one another, C_{10}H_9 formation rates are calculated for 1 and 100 bar, Fig. 6. This is done by using the fastest rates to approximate the chemically activated route ($2\text{CPDyl} \rightarrow \text{N1b} + \text{H}$) and the emission route ($\text{T1b} \rightarrow \text{N1b} + \text{H}$), while estimating the rate of the hydrogen abstraction route as $\text{T1b} + \text{CPDyl} \rightarrow \text{N1b} + \text{CPD}$ using the Reaction Mechanism Generator (RMG) and its corresponding database ($A = 1.1 \cdot 10^{-9} \frac{\text{m}^3}{\text{mol s}}$, $n = 4.3$, $E_a = 48 \frac{\text{kJ}}{\text{mol}}$) [49]. For concentrations, CPDyl is assumed in equilibrium with the $\text{C}_{10}\text{H}_{10}$ isomers since CPDyl is the starting species and the radical recombination reaction is fast. The total concentration of CPDyl + 2 $\text{C}_{10}\text{H}_{10}$ is constrained to $2 \cdot 10^{-9} \frac{\text{mol}}{\text{cm}^3}$ for all conditions to simplify the comparison. This concentration is obtained using simulation of CPD pyrolysis using the POLIMI 1311 mechanism at the low dilution conditions of Djokic et al. at 1000 K [33]. See Table 1 for data used to generate Fig. 6. For high temperatures, the chemically ac-

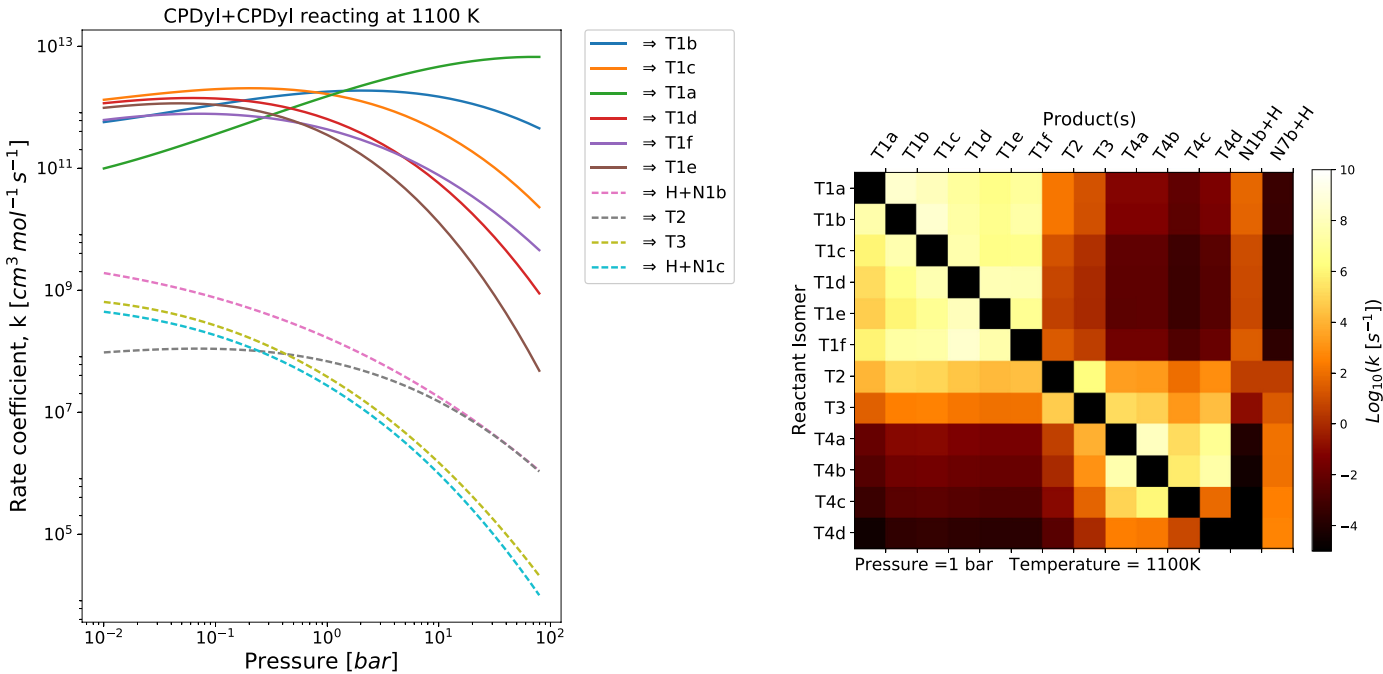


Fig. 5. (A: Left) Computed pressure dependent rate coefficients for the recombination of two CPDyl radicals to form various products in N_2 at 1100 K. The corresponding PES is shown in Figs. 3 and 4. (B: Right) Rate constant matrix depicting rate constant magnitudes for all well-skipping and direct reactions between isomers of the $C_{10}H_{10}$ PES. Reacting species shown on the y axis with the product species along the x axis.

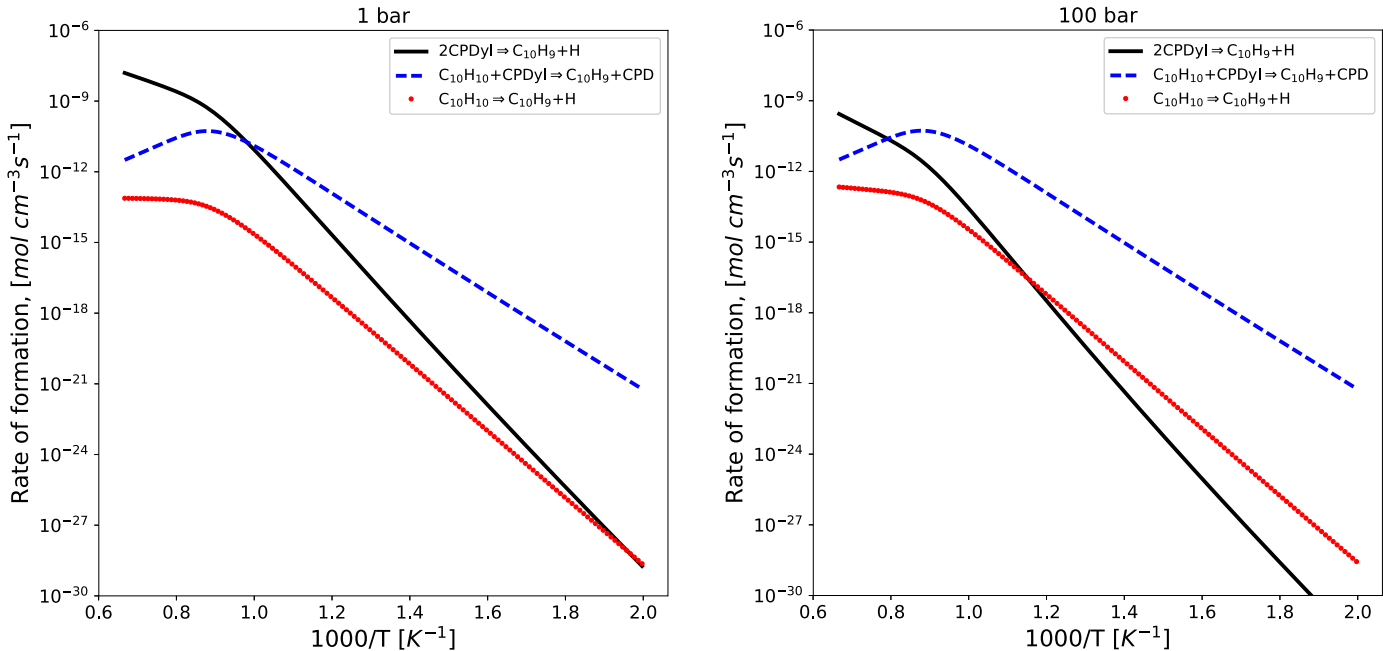


Fig. 6. Rate of $C_{10}H_9$ formation comparison between hydrogen emission from the CPDyl dimer, chemically activated hydrogen emission from the CPDyl + CPDyl reaction, and radical hydrogen abstraction from the CPDyl dimer. At moderate to low temperatures, radical abstraction dominates while at high temperatures the activated path leads formation of $C_{10}H_9$.

Table 1
Selected data set used for the rate analysis of $C_{10}H_9$ formation routes as shown in Fig. 6.

Pressure (bar)	1			100		
Temperature (K)	800	1000	1200	800	1000	1200
k CPDyl + $C_{10}H_{10} \rightarrow C_{10}H_9 + CPD$ ($\frac{cm^3}{mol s}$)	3.5E + 06	3.9E + 07	2.2E + 08	3.5E + 06	3.9E + 07	2.2E + 08
k 2CPDyl $\rightarrow C_{10}H_9 + H$ ($\frac{cm^3}{mol s}$)	2.4E + 06	5.1E + 07	4.5E + 08	3.2E + 03	1.6E + 05	3.0E + 06
k $C_{10}H_{10} \rightarrow C_{10}H_9 + H$ (s^{-1})	9.2E - 10	2.8E - 06	5.0E - 04	1.2E - 09	4.3E - 06	9.8E - 04
[CPDyl] ($\frac{mol}{cm^3}$)	1.0E - 11	4.3E - 10	1.9E - 09	1.0E - 11	4.3E - 10	1.9E - 09
[$C_{10}H_{10}$] ($\frac{mol}{cm^3}$)	9.9E - 10	7.9E - 10	6.0E - 11	9.9E - 10	7.9E - 10	6.0E - 11
[CPDyl] + 2[$C_{10}H_{10}$] ($\frac{mol}{cm^3}$)	2.0E - 09	2.0E - 09	2.0E - 09	2.0E - 09	2.0E - 09	2.0E - 09

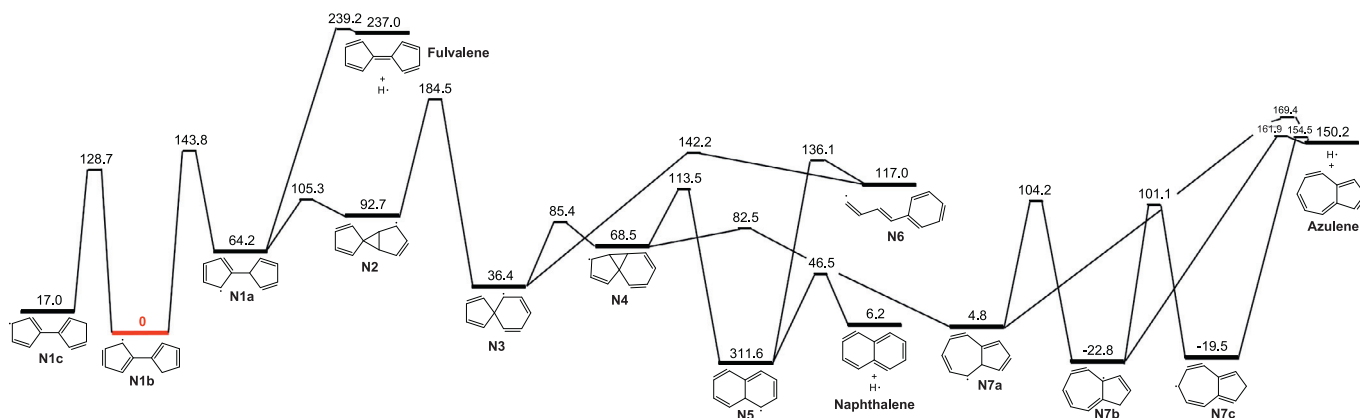


Fig. 7. CBS-QB3 enthalpy diagram $H(0\text{ K})$ (kJ/mol) on the C_{10}H_9 surface leading to the formation of naphthalene and azulene. All enthalpies are relative to component N1b.

tivated route to $\text{C}_{10}\text{H}_9 + \text{H}$ plays a major role. Although $\text{C}_{10}\text{H}_{10}$ is the dominant product of recombination of the two CPDyl radicals, the subsequent thermal decomposition of the $\text{C}_{10}\text{H}_{10}$ isomers to $\text{C}_{10}\text{H}_9 + \text{H}$ is predicted to play essentially no role in the formation of naphthalene. This is because $\text{C}_{10}\text{H}_{10} \rightarrow \text{CPDyl} + \text{CPDyl}$ is energetically favored over $\text{C}_{10}\text{H}_{10} \rightarrow \text{C}_{10}\text{H}_9 + \text{H}$ (Fig. 2).

At 1 bar the radical abstraction route from $\text{C}_{10}\text{H}_{10}$ to C_{10}H_9 becomes the dominant pathway below 1020 K. The crossover temperature rises with pressure to 1230 K at 100 bar. Therefore, radical hydrogen abstraction is particularly relevant at the conditions of pyrolytic experimental studies of CPD. Additionally, the range of relevance is expected to extend to higher temperatures if the full system radical pool is taken into account as possible abstractors rather than just CPDyl as is done here. The marked shifts in slope for each of the three curves are attributed to a shift in equilibrium to favor CPDyl over $\text{C}_{10}\text{H}_{10}$ as seen in Table 1. For temperatures above this point, CPDyl is the major species while $\text{C}_{10}\text{H}_{10}$ becomes the minor species and its concentration begins to fall at an increased rate. A figure showing this trend is included in support-

ing information. An important consequence of the observed importance of radical H abstraction is a shift in net ‘radical’ balance through the formation of naphthalene from CPDyl at these lower pyrolytic temperatures: “Two radicals in, two H atoms out” (when the chemically activated channel dominates) becomes “three radicals in, one H atom out” at lower temperatures, causing potentially a marked reduction in system reactivity.

3.2. C_{10}H_9 PES to azulene and naphthalene

The direct spiran path from the fulvalanyl radical to naphthalene used here (Fig. 7) is consistent with the one originally shown by Melius et al. The surface was subsequently re-calculated at the B3LYP/6-31G(d,p) level of theory by Wang et al. as well as Kislov and Mebel who include this route as part of a larger study [38,39]. In the current study, CBS-QB3 based energies are used. Based on the discussion above, the low-lying C_{10}H_9 species N1b is formed most rapidly. However, H atom scrambling in species N1b is fast and leads to the formation of its isomers N1a and N1c. Species N1c

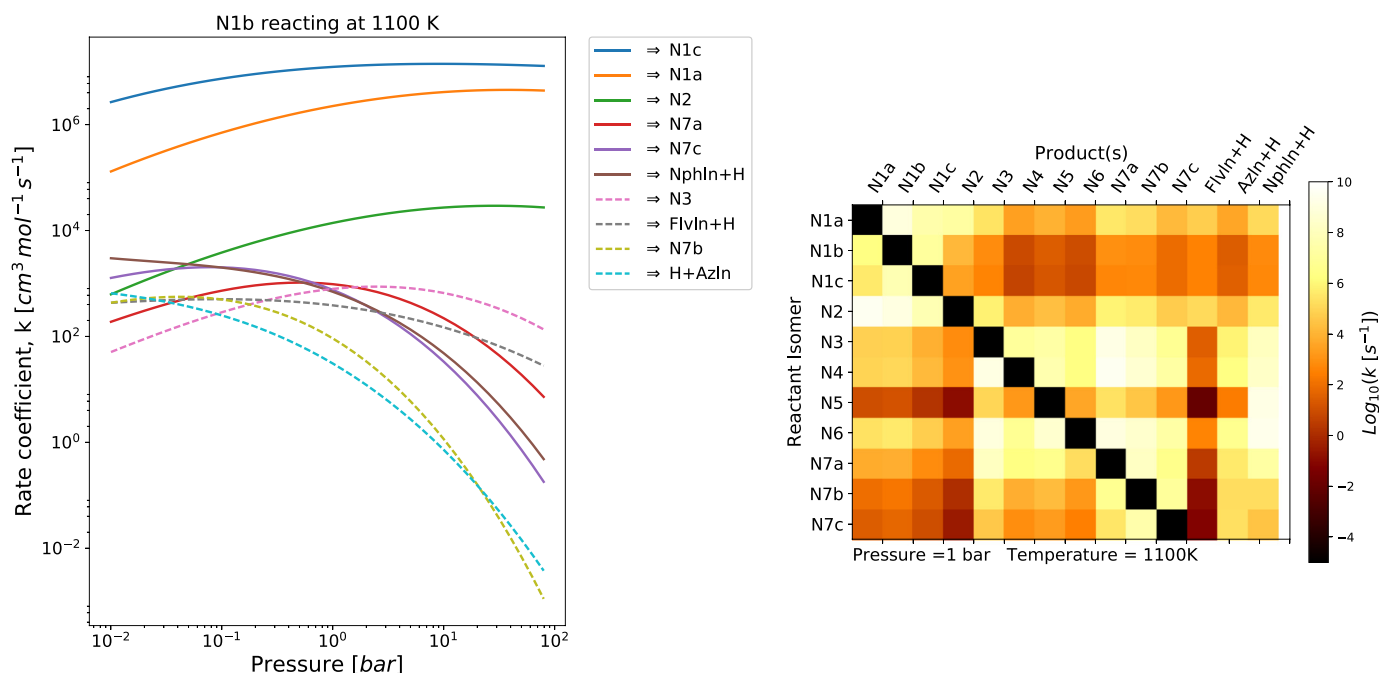


Fig. 8. (A: Left) Computed pressure dependent rate coefficients for the species N1b, the primary entrance channel to the C_{10}H_9 surface to form various products at 1100 K in N_2 . The corresponding PES is shown in Fig. 7. (B: Right) Rate constant matrix for well-skiping and direct reactions among the isomers of the C_{10}H_9 PES. Reacting species shown on the y axis with the product species along the x axis. [Flvln = fulvalene, Azln = azulene, Nphln = naphthalene].

can undergo a 1,3-cycloaddition leading to species **N2**. β -scission of this component leads to the spirane-type species **N3**. This component then reacts via a 1,3-cycloaddition and β -scission to species **N5**, which yields naphthalene by β -scission of a C–H bond.

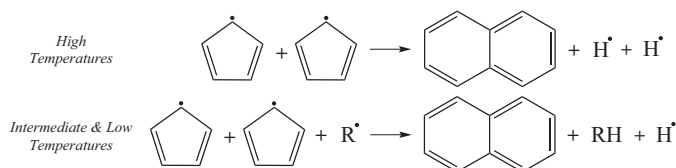
Though the azulane species are not generally observed to be the dominant product on the $C_{10}H_{10}$ PES at the conditions discussed in this study, the Cavallotti et al. publication adds relevance for azulanyl radical conversion to naphthalene in addition to the route from fulvalanyl radicals. Routes between azulene and naphthalene were previously explored by Alder et al. who found two dominant options: the spiran path, and the methylene walk route [44]. Entrance channels for the methylene walk path were observed to be significantly higher than for the spiran route, and thus the methylene walk route has been neglected here. This is in agreement with the expectations of Cavallotti et al. who proposed the spiran route to follow from the azulanyl formation shown in their study. As might be expected, the Alder spiran path merges with the Melius spiran route from fulvalanyl to naphthalene. This occurs at species **N4** and the combined surfaces are shown in Fig. 7.

Considering the $C_{10}H_9$ PES shown in Fig. 7, the pressure-dependent rates shown in Fig. 8 are computed. The primary entrance species, **N1b**, equilibrates with the other fulvalanyl isomers **N1a** and **N1c** on a sub-microsecond timescale. The bottleneck on this PES is the ring opening reaction **N2**→**N3**, passing over a 90 kJ/mol barrier. Interestingly, after passing through species **N2**, the $C_{10}H_9$ species is predicted to pass through the azulanyl species before proceeding on to naphthalene. Since bimolecular reactions are observed to be of relevance for $C_{10}H_{10}$, the lifetime of the $C_{10}H_9$ species is determined by simulating the $C_{10}H_9$ sub-mechanism using CHEMKIN. When starting with an initial concentration of **N1b**, the formation of $C_{10}H_8$ (overwhelmingly naphthalene) follows an exponential curve. This gives $C_{10}H_9$ half-lives of about 20 ms at 800 K, and 20 μ s at 1200 K, for a pressure of 1 bar. At these lifetimes bimolecular reaction rates between $C_{10}H_9$ and other species are expected to be relevant, diverting away from naphthalene formation and on to higher carbon number and other species. For example, in combustion systems $C_{10}H_9 + O_2$ reactions are expected to be especially relevant. Rate based automated mechanism generators such as RMG consider all feasible elementary reactions and will – if necessary – include automatically important bimolecular reactions for $C_{10}H_9$ during mechanism generation.

4. Conclusions

In this study, a pressure-dependence analysis of the CPDyl + CPDyl recombination reaction and subsequent reactions leading to naphthalene is provided. At all low to intermediate temperature conditions (up to 1020 K at 1 bar and 1230 K at 100 bar), naphthalene is mainly formed via a series of reaction steps starting with the formation of $C_{10}H_{10}$ isomers and not through the direct chemically activated reaction $C_5H_5 + C_5H_5 \rightarrow C_{10}H_8 + H + H$. The addition of deeper wells to the $C_{10}H_{10}$ potential energy surface from the Cavallotti et al. work shifts the equilibrium of the cyclopentadienyl recombination reaction more to the right and allows the $C_{10}H_{10}$ species to accumulate more than predicted from the truncated potential energy surface used by Melius et al. and Kislov and Mebel [37,39]. Since C–H scission of any $C_{10}H_{10}$ isomer is highly endothermic, the $C_{10}H_{10}$ species are predicted to accumulate to high concentrations which enable bimolecular hydrogen abstraction reactions to become competitive. $C_{10}H_9$ formation through H abstraction is expected to be the preferred route at pyrolytic conditions at 1 bar and temperatures below 1020 K. H abstraction is predicted to be even more important at higher pressures. The net effect of this analysis is that at low to intermediate temperatures naphthalene formation via CPDyl recombination

consumes three radicals (two CPDyl and one abstracting radical (or atom)) while just one H atom is produced. This is in stark contrast to the simplified scheme ($C_5H_5 + C_5H_5 \rightarrow C_{10}H_8 + H + H$), which implies that naphthalene formation goes along with the conversion of two relatively stable CPDyl radicals to two more reactive H atoms. Therefore the reaction pathways are summarized as:



Isomerizations on the $C_{10}H_9$ surface directly to naphthalene are considered as well as those that pass through the formation of the azulanyl radical. This work indicates that $C_{10}H_9$ intermediates will be collisionally stabilized and live for a period of roughly 20 μ s at 1200 K and 1 bar before decomposing to $C_{10}H_8 + H$. Pressure dependent rate coefficients in CHEMKIN compatible format are given in Supplementary Materials for use in future kinetic modeling studies.

The formation of the first PAHs is a complex process that involves many intermediates and reactions. The growth to larger PAHs, such as anthracene and phenanthrene, would require many more possible intermediates and reactions and might prove too complex to study manually. Automated network generators, such as RMG, and automatic reaction discovery are hence indispensable in order to unravel the important steps in the growth of PAHs [49,73]. Studies such as this one help to provide the integral theoretic backbone on which automatic software operates when predicting more complex chemistries. Future work should try to address the growth to tri- and tetracyclic PAHs through automated means.

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Supplementary materials

- *additionalFigures.pdf* – Contains the additional figures and information mentioned, but not included in the main text. For example the sensitivity to the assumed rates figure is included in this file.
- *chemkin.inp* – Contains a complete, pressure-dependent Chemkin sub-mechanism for the surfaces investigated in the work.
- *speciesGeom.txt* – Includes the optimized geometries for all species, stable and transition states, used in this study.

Supplementary material associated with this article can be found, in the online version, at doi:10.1016/j.combustflame.2017.09.008.

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